

KISANKAVACH

Karnataka

API & Integration Specification

Document control	Value
Document ID	KK-KA-API-001
Version	1.0
Status	Product / Delivery Baseline
Pilot geography	Karnataka
Issue date	18 August 2026
Classification	Controlled - programme stakeholders

Farmer Credit. Benefits. Protection.

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Note: Page references are represented by section numbering to preserve stability across approved revisions.

1. Purpose and Scope

Defines the Karnataka pilot baseline for farmer-centred Kisan Credit Card enablement, benefit visibility, consent-driven data exchange, grievance handling, fraud protection and programme oversight.

In scope: farmer mobile experience, assisted service, participating-bank workflow, Government command centre, authorised integrations and operational reporting.

2. Guiding Principles

Kannada-first and low-literacy friendly; reuse verified information; explicit purpose-bound consent; least privilege; interoperability through adapters; configurable workflows and auditability.

The bank remains the final credit decision-maker. Government-originated and bank-originated data retain their source and provenance.

3. API Standards

HTTPS REST/JSON under /api/v1; UTF-8; ISO-8601 UTC timestamps; Indian currency in integer paise or documented decimal; stable enums; OpenAPI contract; correlation ID on every request; no sensitive data in URLs.

4. Authentication and Headers

Header	Requirement
Authorization	Bearer token for protected calls
X-Correlation-ID	Client supplied or generated and returned
Idempotency-Key	Required for submission and payment-like mutations
Content-Type	application/json or approved multipart upload
X-Consent-ID	Required when the operation relies on farmer consent

5. Core Endpoints

Method / path	Purpose	Key response
POST /auth/otp/request	Request OTP	challenge_id, expiry
POST /auth/otp/verify	Verify and create session	access/refresh token metadata
GET /farmers/me	Get aggregated profile	profile plus provenance
POST /consents	Record decision	consent_id and status
POST /kcc/applications	Create draft	application_id/public_ref
PATCH /kcc/applications/{id}	Save validated draft step	version and validation result
POST /kcc/applications/{id}/submit	Idempotent submission	status and submitted_at
GET /kcc/applications/{id}/timeline	Farmer-safe status history	current state and actions
POST /documents/uploads	Initiate secure upload	upload token/object reference
POST /grievances	Create grievance	case reference and due date

6. Canonical Payload

```
{
  "applicationId": "...",
  "farmerId": "...",
  "bank": {"bankId": "...", "branchId": "..."},
  "requestedAmount": {"currency": "INR", "value": "200000.00"},
  "consentId": "...",
  "version": 4
}
```

7. Errors and Idempotency

HTTP	Code	Behaviour
400	VALIDATION_FAILED	Field errors; do not retry unchanged
401/403	AUTH_REQUIRED / FORBIDDEN	Reauthenticate or deny without leakage
409	STATE_CONFLICT	Refresh current resource/version
422	BUSINESS_RULE_FAILED	Explain allowed corrective action
429	RATE_LIMITED	Retry after advised interval
503	DEPENDENCY_UNAVAILABLE	Safe retry with backoff

An idempotency record binds key, caller, operation and request hash; repeats return the original outcome, while a different payload using the same key is rejected.

8. FRUITS and Government Integration

Adapter maps authorised farmer, land and crop responses into canonical objects. Every response records source system, source reference, retrieval timestamp, verification status and consent ID. Cache only within approved purpose and freshness rules.

9. Bank Integration Contract

Common operations: submitApplication, getApplicationStatus, requestAdditionalDocument, updateSanctionStatus, updateRejectionStatus, updateDisbursementStatus and getRenewalStatus. Supported modes are real-time REST, secure asynchronous API, secure batch/file exchange and controlled portal workflow.

10. Webhooks and Events

Signed callbacks include event_id, event_type, occurred_at, resource reference, sequence/version and correlation ID. Consumers verify signature, timestamp and replay window, persist receipt, process idempotently and return promptly.

11. Operational Controls

Per-partner credentials and limits; timeouts; exponential backoff with jitter; circuit breakers; dead-letter queues; dashboards; reconciliation reports; certificate/key rotation; sandbox certification and production cutover checklist.

End of controlled baseline